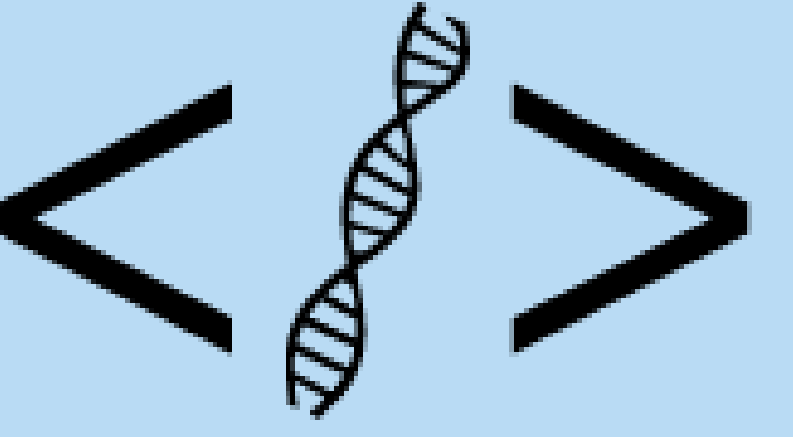


Discovery of candidate cis-regulatory elements from single-nucleus multi-omics data with CreFinder



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INTRODUCTION

Cis-regulatory elements (CREs) contain the regulatory logic that precisely governs the expression of individual genes in specific cell types, playing a key role in driving cell differentiation and maintaining cellular function. Variations in CREs can lead to gene dysregulation, contributing to disease development, while also serving as a source of essential evolutionary adaptations. Due to their gene- and cell-type-specific nature, deciphering CRE regulatory mechanisms across different cell types can be difficult.

Here, we introduce a novel framework in python, CreFinder, that leverages on paired single-nucleus multi-omics data and the context likelihood of relatedness (CLR) algorithm to infer whole genome, cell-type-specific candidate CRE (cCRE) regulation on genes¹.

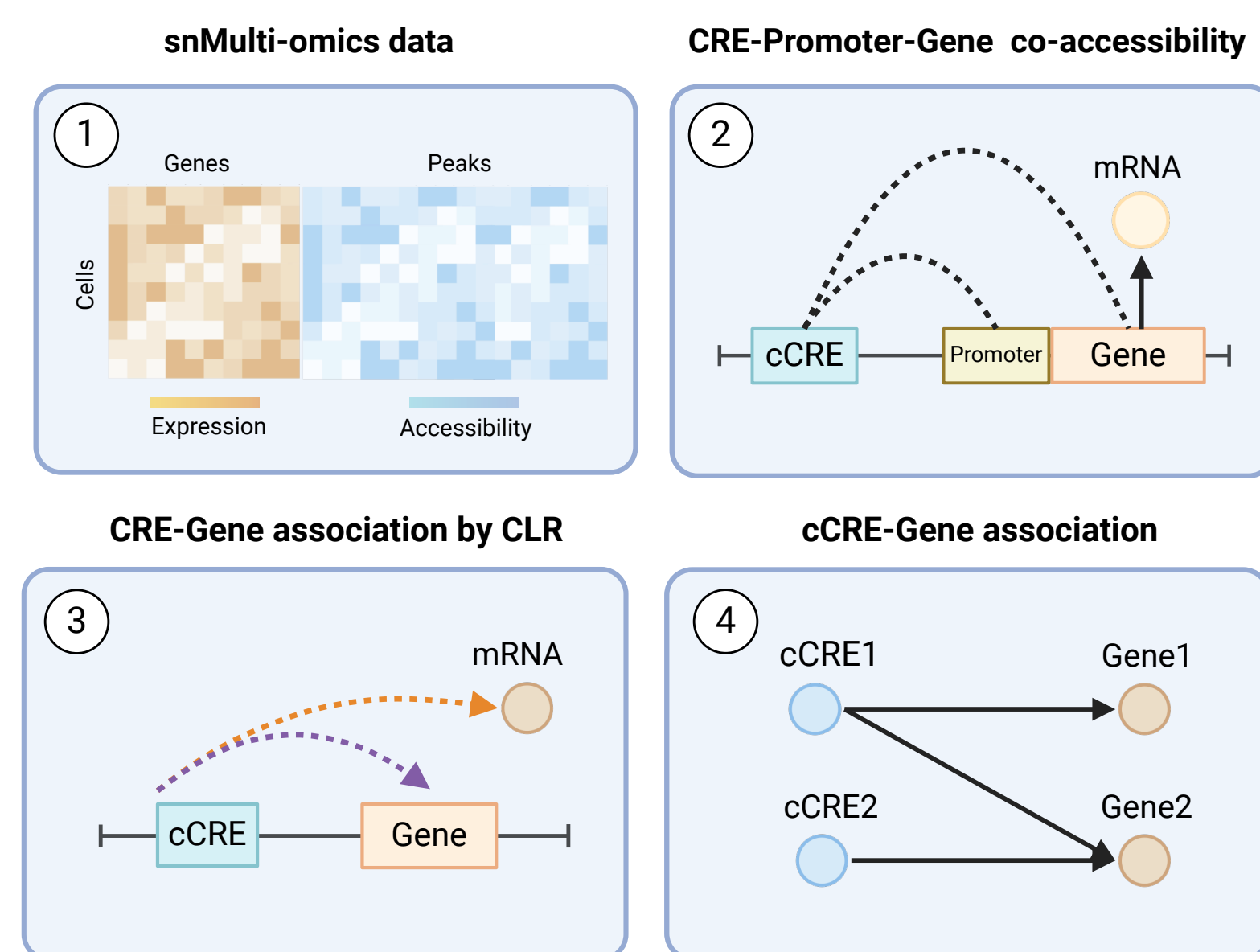


Figure 1: Workflow. CreFinder takes (1) paired single-nucleus RNA-seq (scRNA-seq) and single-cell ATAC-seq (scATAC-seq) data with (2) CRE-Promoter-Gene co-accessibility assumption and (3) CLR algorithm to (4) infer cCRE-gene link.

DATA PRE-PROCESSING

When a gene is actively transcribed, its CREs loop over to physically contact with the gene promoter, initiating transcription. During this process, the CREs, promoter, and gene body become accessible, leading to a rapid increase in the gene's mRNA count. In pre-processing, we select genes that exhibit shared co-accessibility and expression on a per-cell basis to study CRE regulation.

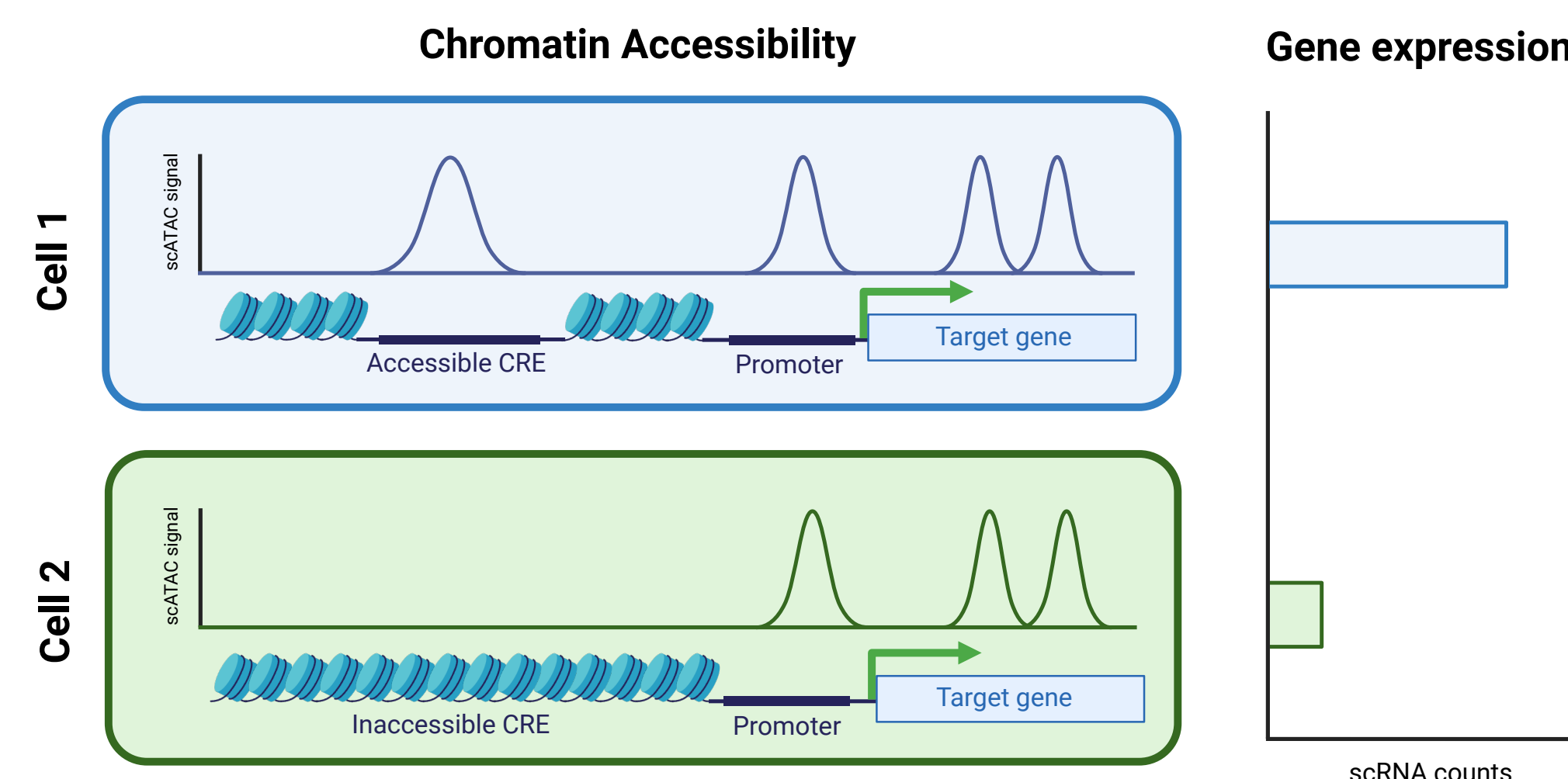


Figure 2: A toy example. Cell 1 and Cell 2 both have accessible promoter and target gene body scATAC-seq peaks. However Cell 1 has accessible CRE peak which drives a higher target gene expression.

METHODS

To integrate information across multi-modal data, we employ the CLR algorithm to associate epigenetics accessibility with target gene expression. The CLR algorithm uses mutual information (MI) to systematically identify highly consistent patterns across diverse cell conditions. By leveraging whole-genome information, it performs background correction by calculating the statistical significance of a cCRE-gene pair's MI score within its genomic context. This approach is not constrained by strict distance boundaries nor reliant on epigenetic biomarker databases.

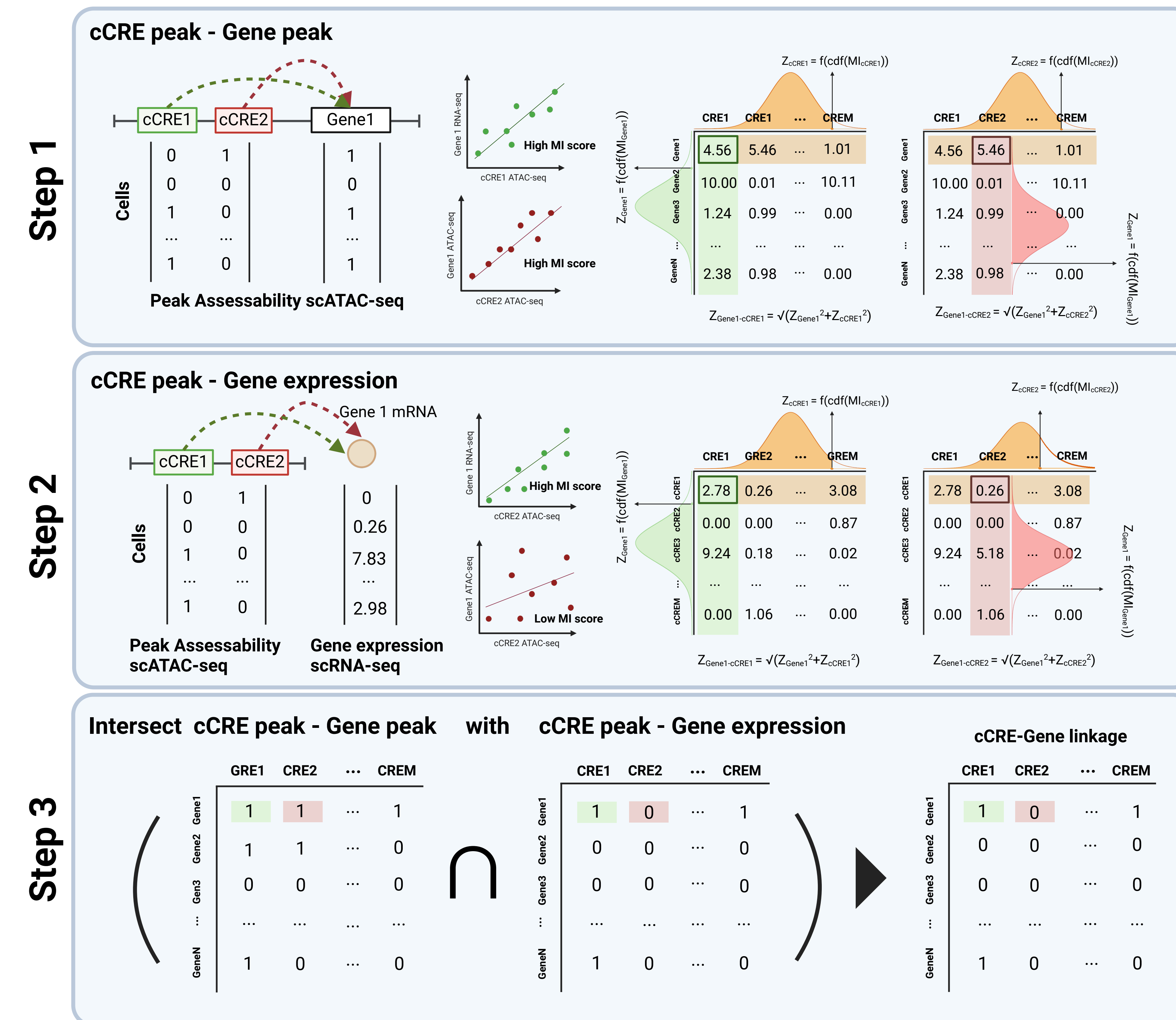


Figure 3: CreFinder Workflow. **Step 1:** Infer associations between cCRE peaks and gene body peaks using the CLR algorithm. cCRE1 and cCRE2 peaks both have high MI association to Gene1 peak. **Step 2:** Infer conditional CLR associations between the selected cCRE peaks and gene expressions. cCRE1 peak accessibility shows increasing association with Gene1 gene expression while cCRE2 does not. In both steps, the CLR algorithm first computes the MI score between two variables across all cells. It then calculates the z-scores for regulatory interactions based on the MI scores distribution of each cCRE (Z_{cCRE1}) and each gene (Z_{Gene1}), and combines the two Z-scores. **Step 3:** Intersect the z-score results from Step 1 and Step 2. cCRE1 is predicted to interact with Gene1.

The core of CreFinder involves three main steps. **Step 1**, CreFinder infers CLR associations between cCRE peaks and gene peaks, based on the assumption that CRE peaks and gene peaks are both accessible during active transcription. User can set z-score threshold to select significant association for downstream analysis. A pair of cCRE and gene is predicted to interact if the MI score is above some set threshold. **Second step**, CreFinder infers conditional CLR associations between selected cCRE peaks from step 1 and gene expression, leveraging the knowledge that the opening or closing of a regulatory CRE will affect gene expression. Another z-score threshold can be set here. **Third step**, CreFinder intersects the two association results to select final cCRE-Gene links.

In **Figure 3**, we demonstrated the enhancer effect of cCRE1 on Gene1. It's also possible for a CRE to exert suppressive effect on gene expression, acting as silencers. We can calculate a simple Pearson correlation between predicted cCRE-Gene pairs to identify their activities. A positive correlation indicates an enhancer effect, while a negative correlation suggests a silencer effect.

RESULTS & CONCLUSION

To verify CreFinder's ability to capture biologically functional CREs, we applied it to predict cCRE-gene linkages using peripheral blood mononuclear cells (PBMCs) data from a healthy 25-year-old female, obtained from 10x Genomics². We first performed quality control with Scanpy to remove low-quality cells and retain only high-quality genes and peaks.

Next, we applied CreFinder's data pre-processing pipeline to filter out genes that do not have accessible promoters and gene bodies and do not show expression on a per-cell basis. During this process, all non-promoter and non-gene body peaks were labeled as CREs. Ultimately, gene body peaks and promoter peaks for the same gene were combined into a single peak, and all ATAC-seq peaks were binarized. The scRNA-seq data was normalized on a chromosome basis.

Since CREs are highly cell-type specific, we chose to focus on B cells. In total, we obtained 179 B cells with 2284 genes that demonstrated co-accessibility and gene expression. A total of 60630 scATAC peaks were used as input. After applying CreFinder on all 23 chromosomes, using a z-score threshold of 3 as the minimum for significant linkage for both cCRE peak-gene peak and cCRE peak-gene expression, a total of 9449 linkage predictions were made.

We then assessed the accuracy of CreFinder's ability to capture possible CREs by overlapping the predicted cCREs with B cell specific cCREs from the ENCODE database, which contains 1063877 peaks³. We identified an average of 95.9% overlap of predicted cCREs across all chromosomes.

These results demonstrate that CreFinder is effective in identifying biologically relevant CRE to gene linkages. The high overlap with the ENCODE database supports the accuracy and reliability of our predictions. Therefore, CreFinder can serve as a valuable tool for exploring cell-type-specific regulatory mechanisms from single cell multi-omics data, potentially aiding in the understanding of gene regulation in health and disease.

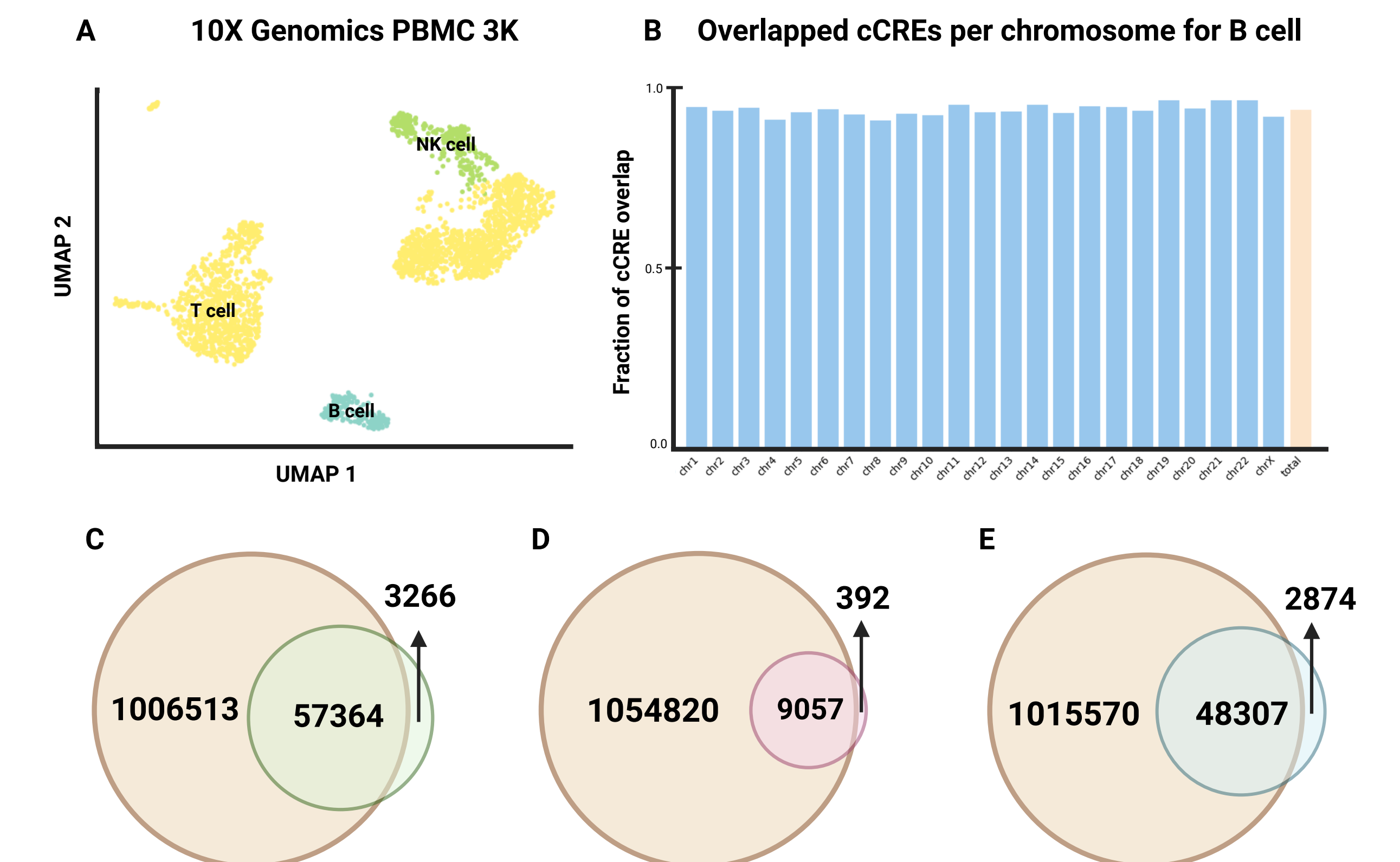


Figure 4: Verification Data and Overlap Fractions for PBMC. (A) UMAP visualization of the PBMC data, showing clusters of T cells, B cells, and NK cells. (B) Overlap fraction of B cell specific cCREs between the ENCODE database and those predicted by CreFinder for each chromosome and in total. (C)-(E) Venn diagrams illustrating the number of overlapped cCREs between (C) ENCODE and all identified peaks, (D) ENCODE and CreFinder-predicted peaks, and (E) ENCODE and non-CreFinder peaks.

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